

SHUBHAM KUMAR DAS

+91-8369899530 | bt23csa006@iiitn.ac.in | [linkedin.com/in/shubham-kumar-das](https://www.linkedin.com/in/shubham-kumar-das) | github.com/SKD151105

EDUCATION

IIIT Nagpur | *B.Tech CSE (AI/ML Specialization)* | **CGPA:** 7.54/10.0 **CBSE:** XII: 95.8% | X: 99.0%

TECHNICAL SKILLS

Languages: JavaScript (ES6+), C/C++, Python

Backend Development: Node.js, Express.js, RESTful API Design, JWT Authentication, Payment Integration (Stripe)

Frontend Development: React.js, Client-Side Routing, State Management (Hooks, Context), API Integration

Databases: MongoDB, PostgreSQL, Mongoose, Redis, Schema Design, Indexed Queries

Security & Best Practices: Role-Based Access Control (RBAC), Rate Limiting, Input Validation, Error Handling

AI/ML Fundamentals: Supervised/Unsupervised Learning, Neural Networks & Deep Learning, Computer Vision

Tools & Platforms: Git, GitHub, Linux, Postman, Vercel, Render, MongoDB Atlas

Competitive Programming: CodeChef (2-Star) | LeetCode (1650+ Rating)

PROJECTS

OmniSocial — Full-Stack Social Platform | *Node.js, Express, MongoDB, React, JWT, Socket.io*

- Built a full-stack social media platform with secure JWT-based authentication using access and refresh tokens
- Implemented real-time private messaging and notifications via Socket.io with event-driven architecture
- Designed backend using layered architecture (Controller-Service-Repository-Model) for separation of concerns
- Modeled relational data (Users, Posts, Comments, Messages) with Mongoose schemas and performance indexing
- Added rate limiting, centralized error handling, and request validation middleware for API reliability and security

PulseCheck — Uptime Monitoring SaaS | *Node.js, Express, MongoDB, Redis, REST APIs*

- Built backend-focused uptime monitoring SaaS enabling users to register websites and APIs for periodic health checks
- Developed monitoring engine to perform scheduled HTTP checks, track uptime status, and record response latency
- Designed background job processing system for executing periodic checks and storing comprehensive logs
- Enforced subscription-based feature limits (free vs. paid) at the service layer using Redis-backed rate limiting

ShortLink — Scalable URL Redirection Backend 🚀 *Node.js, Express, MongoDB, Redis, Zod*

- Engineered a **short link generation and URL redirection** system using **repository-pattern architecture** with thin controllers and isolated business logic
- Benchmarked **100 concurrent connections** via autocannon; achieved **1,365 req/sec** at **73 ms avg latency** (p95: 138 ms)
- Applied Redis **positive and negative caching**, boosting throughput **12x** (valid URLs) and **9x** (invalid lookups) while cutting latency by **92%**
- Built **collision-resistant short code generation** with unique-constraint retries and adaptive length expansion
- Added **non-blocking click analytics** to capture IP and user-agent metadata without impacting redirect latency

ACHIEVEMENTS

2nd Place — IIITN Hackathon Solve-a-Thon (**National-Level**)

1st Place — Cypher Chronicles (Public Speaking & Presentation), IIITN

2nd Place — InFacto Oxford-Style Debate Competition, IIITN

JEE Mains — **98.298 Percentile** (AIR 19,918; **Top 1.7%** among 11.7L candidates)